

2022 Middle Mathematical Contest in Modeling (MidMCM), Team Control Number: 13575 Problem Chosen: C

1 | Executive Summary

The problem that we chose was Problem C. This problem requires us to create a schedule for a group of friends in a theme park. However, before we proceed, we must give a bit of background as to what this theme park offers. There are many attractions and sites that can be found in the Polygon Park, such as the different thrilling roller coasters and flumes, the fun arcade and carnival games, awesome shows from acrobatics to magic, and even delicious food that may be bought in the restaurants or food stands (hotdogs, iced cream). The problem stems from here, as though not only would we have to create a good park experience given the waiting times, and time schedules given out by the programs, but also conform quite well with the requests of all of the friend group.

To begin the task of scheduling, we, as a team, familiarized ourselves with what exactly the theme park can offer, and what each of the friends really desire from this trip to the theme park; we went as far as to really sympathize and go into the friend group's shoes as to what exactly *we* would want in a theme park vacation (as normal teenagers). This starting point allowed us to see the problem in a much wider scope, led us to understand exactly what we wanted (an optimized schedule), and made us capable of solving the problem with our model (solution) that we shall present in the next few pages.

Our solution mainly consists of us making smaller, sub-problems (creating individual schedules), then simply joining all of these schedules together to create the final solution. Due to the nature of having only one person for a schedule, it is far easier to create a schedule for each of them (which we exhibited in Section 5.1), and can be done simply using logical deduction. We started the solution by first creating a schedule of all of the friends, then getting the intersections in the schedules, together with all of the activities that they agree to do. From there, we get a schedule that has a surprising amount of events. That is the first half of our model: getting the general consensus.

The last, far more challenging half of the problem. Since the schedules are clashing against each other (for instance, Ishmael loves rides but Ming has motion-sickness), we eventually found ways to be able to compromise the schedules, ranging from diverting the schedules, to even making some do activities they do not really favor, for the common good. This is quite hard to summarize whilst going in-depth, therefore I shall save the in-depth explanations for the solution which may be found at Section 5.

2 | Table of Contents

1 Executive Summary	1
2 Table of Contents	2
3 Introduction	3
3.1 Background and Problem Statement	3
3.2 Interpretation of Information Given	3
4 Assumptions and Definitions	6
4.1 Assumptions and Justifications	6
4.2 Definition of Terms and Variables	7
5 Model and Solution	9
5.1 Gathering Each Schedule	9
5.2 General Plan of Action for Final Schedule	12
5.3 Final Schedule	13
5.3.1 List of factors for enjoyment in the final schedule	13
5.3.2 Combined schedule of Ishmael and Freya	14
5.3.3 Schedule of Ming and Karine During the Split	15
6 Analysis	16
6.1 Strengths	16
6.2 Limitations	16
6.3 Reflection	16
7 Conclusion	17
8 Letter to Group	18
9 References	20

3 | Introduction

3.1 Background and Problem Statement

We were tasked by a group of friends to make a schedule for their activities during their trip to a small amusement park, Polygon Paradise Park. This schedule must keep in mind all of the friends' interests and needs. In the schedule, we have to account for walking time, eating, and going to the restroom or water closet. The group will only allow a split-up into 2 groups of 2 people each for a maximum of 4 hours.

Each person is looking for something to do at Polygon Paradise Park. The theme park offers rides, games, and food. According to what the group of friends has told us, we can get an idea of how they want to spend their day. Based on this, we created a model for their schedule with what we guessed would be an ideal experience for them.

The objective of our paper is to create a schedule that maximizes the overall enjoyment of the 4 friends. With the general likes and dislikes we created a schedule to make everyone as happy as possible without creating situations that would make some individuals happy.

3.2 Interpretation of Information Given

About the preferences of the group:

- Ming loves to play games and win prizes. Ming gets motion sick on fast moving rides especially after eating.
- Ishmael loves roller coasters and wants to ride as many of them as possible during the day. Ishmael also likes performing card tricks and is interested in seeing the magic show.
- Karine is very impatient and hates waiting in long lines. Karine wants to attend the circus acrobatics show. Karine does not eat hot dogs.
- Freya likes to spend time on water rides to cool off. Freya likes to go with the flow and is happy spending time with friends. Freya wants to make sure she gets a souvenir to remember the trip.

About the Ideal Experiences:

- It is said that **Ming** would like to be in the arcade for a lot of the time, winning prizes and playing games. This implies that Ming generally likes playing or doing more active activities such as playing bumper cars (pentagon bumpers) rather than sitting at a ride

(also due to the fact Ming has motion sickness). Ming would definitely not want to go on any of the rides as much as possible, and definitely not right after eating.

- **Ishmael** would simply enjoy going to as many roller coasters as possible, and that kind of generalizes to wanting to go to as many rides as possible (and the most variety of them for a new thrilling experience). “As many as possible” also seems to lead to a bit of similarity to Karine’s problem, as though they’d want the least waiting time as well. Ishmael would also want to watch the magic show as a magic fanatic.
- **Karine** would just want to watch the circus acrobatics show. Otherwise, they do not really seem too much of a fan of amusement parks in general, especially with their pet peeve of long lines (as an impatient person). This would either mean going to activities that do not have any lines indicated, or there should be a time slot where there are barely (or zero) minutes of waiting time.
- Based on the description, as long as **Freya** gets to do the water activities, they would be completely fine with doing anything that their friends are doing. Freya seems to match far more with Ishmael than the two other friends in terms of the amusement parks because Ishmael also loves going on rides and generally has no necessary explicit restrictions.

About Polygon Paradise Park:

- Open from 9 AM to 9 PM
- There are 10 rides of varying thrill levels and duration
 - Heptagon Coaster: Extreme Thrill 2.5 min
 - Decagon Coaster: Extreme Thrill 3.0 min
 - Rhombus Ride: Extreme Thrill 4.0 min
 - Hexa-Swings: Medium Thrill 2.0 min
 - Pentagon Bumpers: Medium Thrill 5.0 min
 - Square Scramble: Medium Thrill 2.5 min
 - Concave Train: Basic 15.0 min
 - Starburst Carousel: Basic 3.0 min
 - Dodecagon Rapids: Water 10.0 min
 - Octagon Flume: Water 5.0 min

- These are wait times for each ride at a given time in the day.

Ride Wait Times (Minutes)

Time	Concave Train	Decagon Coaster	Dodecagon Rapids	Heptagon Coaster	Hexa-Swings	Octagon Flume	Pentagon Bumpers	Rhombus Ride	Square Scramble	Star Burst Carousel
9:00 AM	0	0	0	0	0	0	0	0	0	0
10:00 AM	5	15	0	10	0	10	0	5	0	10
11:00 AM	10	30	5	20	5	10	5	10	5	15
12:00 PM	15	45	10	30	10	10	5	15	10	15
1:00 PM	10	60	15	40	10	30	5	10	10	10
2:00 PM	10	75	20	40	15	60	10	10	10	5
3:00 PM	15	75	15	30	15	60	10	15	10	5
4:00 PM	15	60	10	30	10	60	10	10	10	5
5:00 PM	15	45	5	20	10	30	10	10	10	5
6:00 PM	5	30	0	20	5	10	5	5	5	5
7:00 PM	5	30	0	10	0	10	0	5	5	5
8:00 PM	5	15	0	10	0	0	0	5	5	5
9:00 PM	5	0	0	10	0	0	0	5	0	0

- There is a Trapezoid show with the schedule attached below.

Trapezoid Show Schedule

Time	Show	Duration
10 am	Circus Acrobatics	1 hour
12 pm	Circus Acrobatics	1 hour
2 pm	Magic Show	1 hour
4 pm	Circus Acrobatics	1 hour
6 pm	Magic Show	1 hour
8 pm	Firework Show	1 hour

- There are 3 food options, 2 of which are takeaway and 1 of which is sit down.
 - Hot Dog Stand: Take Away 10:00 AM – 9:00 PM
 - Ice Cream Cart: Take Away 11:00 AM – 9:00 PM
 - Triangle Restaurant: Sit Down 9:00 AM – 8:00 PM

4 | Assumptions and Definitions

4.1 Assumptions and Justifications

For the Amusement Park:

- 1) Assumption: *Each person in the group will have to use the restroom at least 4 times and eat at least twice.*

Justification: *We can say that they have already eaten breakfast since they arrive at 9:00am, so the only meals left are Lunch and Dinner, and possibly snacks. The average person goes to the toilet about 4-10 times a day, according to healthpartners.com.*

- 2) Assumption: *These are a completely normal group of friends, meaning that they follow a general bandwagon. For instance, These people would want to try out everything except the things they hate.*

Justification: *As there are no other descriptions provided by the friends, we would assume that this friend group is in the teenager age range as they are going to a theme park alone. As teenagers, they would usually conform to the usual activities that most people would partake in, ergo our assumption.*

- 3) Assumption: *The group of friends would spend all the time from 9 am - 9 pm in the park; meaning that they are free to do an activity anytime in that time window.*

Justification: *As they are going to the theme park for one day, it would be best if they could maximize the entire 9AM-9PM in the theme park, though still with breaks but still generally in the theme park for the entire duration of the day-to-night.*

- 4) Assumption: *We will just set all the walking times to 7 minutes with a little bit tolerance as this is the longest time it will possibly take for the group to walk from a place to another place (refer to the Polygon Paradise map below).*

Justification: *This way, it would be far easier to create the approximations needed to create a generally-fitting schedule for the group of friends.*

- 5) Assumption: *We will also assume that the weather conditions during the day they go to the park are ideal. The average room temperature is 20 degrees celsius so we will assume that the park is this temperature all throughout the day.*

Justification: *This would also help us make it easier to approximate the different times in the schedule without any interruptions that could arise if the weather is*

bad or if anything wrong happens in the theme park.

- 6) Assumption: *Water rides in the park fall under extreme thrill rides.*

Justification: *It would be easier to account for Ming's motion-sickness in rides.*

- 7) Assumption: *The Triangle Restaurant accepts takeaways for leftovers.*

Justification: *Most sit-in restaurants accept taking leftovers to eat for later. This would allow us to keep a slightly tighter constraint for eating lunch or dinner, since our model mostly assumes that they eat at a normal pace.*

Assumptions on Favorable Conditions (based on description):

- Favorable
 - *Ming wants to go to the arcade*
 - *Stated in the problem.*
 - *Ishmael wants to go to as many roller coasters*
 - *Stated in the problem.*
 - *Karine wants to go to the circus acrobatics show*
 - *Stated in the problem.*
 - *Freya wants to go on water rides*
 - *Stated in the problem.*
 - *Freya wants as much time as one group*
 - *Stated in the problem.*
- Not Favorable
 - *Ming does not want to go on a medium / extreme ride*
 - *This is simply due to their motion-sickness.*
 - *Ming does not want to go on any ride right after eating*
 - *This is also due to the motion-sickness.*
 - *Karine does not want to wait in long lines*
 - *This is part of the problem statement.*
 - *Karine does not want to eat hot dogs*
 - *This is part of the problem statement.*

4.2 Definition of Terms and Variables

Notation	Definition	Notes
n_r	Number of unique rides ridden.	This is something we would want to <u>maximize</u> (Ishmael)
t_{wi}	Total waiting time on i^{th} block on schedule.	This is something we would want to <u>minimize</u> (Karine)
t_{ri}	Total riding time of i^{th} rollercoaster.	This is something that is constant based on the ride
t_s	Total time separated from the group.	This is something we would want to <u>minimize</u> (Freya)
$t_g = 7 + \varepsilon$	Time spent on walking to a destination (grace period), according to healthline.com and other sources.	This is a (supposedly) controlled variable, equal to <u>7 mins</u> . However this may vary slightly by some small amount of time ε .
e_I, e_M, e_F, e_K	Enjoyment of Ishmael, Ming, Freya, and Karine respectively.	This is more of a short way or more mathematical way of representing the enjoyment of a certain group of friends.

Term	Definition
Sites / Attraction	The certian spot in the Amusement Park.
Optimize	Create a schedule that would best suit the entire friend group.
ε	A "small" value.
Intersection	A time slot that all the groups share.
Consensus	An activity that all of the group wants to do, but not necessarily same time.

5 | Model and Solution

5.1 Gathering Each Schedule

Our plan to ensure maximum enjoyment of each individual and total enjoyment was to create separate schedules for each person in the group. However, according to calendar.com's blog, we must remind ourselves that this is a vacation and not a speedrun as to being able to get all of the checkmarks of the friends done; therefore, we would finetune the schedule to have a lot of breaks in between the activities. Using their separate schedules, we can take equal amounts of activities from each person and fit it into the 12-hour time frame (9 AM - 9 PM). This would allow us to have a more-or-less fair schedule that would be favorable to all of the group of friends. The methodology of combining all of these schedules would be the main part of the solution.

In the following ideal schedules, any time jumps will constitute a break for snacks, going to the restroom (WC), and/or any activity they wish. Instead of enforcing the walk time for 7 minutes (t_g), we instead put an allowance for each schedule block that will make time for slow walkers or fast walkers ($t_g + \epsilon$). There are huge chunks of time without scheduled activities but this is because we are simply highlighting the important activities that should be done in the individual schedules. We will use this to formulate a template for the joint/final schedule and *then* fill out the blanks. Note that the following would be the schedules for each of the friends in the friend group, and would be simply justified based on their description.

5.1.1 Ishmael's Ideal Schedule

Timeslot	Activity	Notes
9:00am - 11:30am	A huge chunk of the rides (Hepta, Deca, Flume, Star, Square, Rhombus, Concave respectively)	$t_g + 2(2.5) + t_g + 2(3) + t_g + 2(5) + t_g + 2(3) + t_g + 2(2.5) + t_g + 2(5 + 4) + t_g + (5 + 10 + 15 + 15) \approx 144 \text{ minutes} \approx 150 \text{ minutes}$ Note that the format of each would be $t_g + 2t_{ri}$, the order indicated in the activity schedule. This is due to the fact that Ishmael, as a fan of roller coasters, would like to ride a roller coaster not only once.
11:30am - 12:50pm	Hotdog Takeaway	Ishmael wants to eat at a takeaway so that he will have more time to go on rides.

12:50pm - 2:00pm	Games	
2:00pm - 3:00pm	Magic Show	Ishmael likes magic shows, therefore it must be in his ideal schedule.
3:00pm - 4:00pm	Games	
4:00pm - 5:00pm	Circus Acrobatics	As aligning with the bandwagon, they would like to see or watch something new.
5:00pm - 6:00pm	Heptagon Coaster	$t_g + 2(20+2.5)$
6:00pm - 7:00pm	Another Magic Show	Since Ishmael likes magic, it would be nice if he would repeat it in his time schedule.
7:00pm - 8:00pm	Finishing of the rides (Rapids, Hexa, Penta respectively)	$2(10) + t_g + 2(2) + t_g + 2(5) + t_g$ Note that the format of each would be $2t_{ri} + t_g$, the order indicated in the activity schedule.
8:00pm - 9:00pm	Fireworks	Aligning with bandwagon, they would want to try out new activities, and one activity that even we'd want to try out is watching the fireworks.

5.1.2 Ming's Ideal Schedule

Timeslot	Activity	Notes
9:00am - 10:40am	All of the rides only once	$t_g + 10 + t_g + 5 + t_g + 3 + t_g + 15 + t_g + 2 + t_g + 5 + t_g + 2.5$ These are all of the rides except the extreme ones, as though even if Ming has motion-sickness they are still teens who want to conform to the norm, and therefore would at least experience the rides once.
10:40am - 11:30am	Games	
2:00pm - 3:00pm	Triangle Restaurant	

4:00pm - 5:00pm	Circus Acrobatics	
6:00pm - 7:00pm	Magic Show	
8:00pm - 9:00pm	Fireworks	

5.1.3 Karine's Ideal Schedule

Timeslot	Activity	Notes
9:00am - 9:50am	A huge chunk of the rides (Hepta, Deca, Flume, Star, Square, Rhombus, Concave respectively)	Dodecagon Rapids, Hexa-Swings, Pentagon Bumpers, Square Scramble (these 4 because they don't have waiting times at 10:00am) = 19.5 minutes $19.5 + 28 = 47.5 \text{ minutes} \approx 50 \text{ minutes}$ Note that the format of each calculation for time would be $t_{ri} + t_g$, with i being based on the order indicated in the activity schedule.
2:00pm - 3:00pm	Triangle Restaurant	Karine can't eat hotdogs.
4:00pm - 5:00pm	Circus Acrobatics	
6:00pm - 7:00pm	Magic Show	
8:00pm - 9:00pm	Fireworks	

5.1.4 Freya's Ideal Schedule

Timeslot	Activity	
9:00am - 10:40am	All the rides	$t_g + 10 + t_g + 5 + t_g + 3 + t_g + 15 + t_g + 2 + t_g + 5 + t_g + 2.5$ Note that the format of each calculation for time would be $t_g + t_{ri}$, with i being based on the order indicated in the activity schedule.
2:00pm - 3:00pm	Triangle Restaurant	

4:00pm - 5:00pm	Circus Acrobatics	
6:00pm - 7:00pm	Magic Show	
7:30pm - 8:00pm	Gift Shop Souvenirs	
8:00pm - 9:00pm	Fireworks	

5.2 General Plan of Action for Final Schedule

We ended up creating some type of heuristic that generalizes to most types of schedules. This would be able to mostly create a schedule based on other people's ideal schedules, which could be generally trivial to be able to formulate, as we have shown in 5.1. This is due to the nature of simply being able to satisfy very straightforward and direct descriptions that do not necessarily have time constraints (aside from waiting times), and is more of an "as long as we are able to do an activity" or "do not do this certain activity."

From those individual schedules, we would finally create our final, joint schedule. We realized that getting the intersection of all of the schedules would be generally optimal for a lot of schedules. This is to minimize t_s , as we may simply take all of the intersections of all of the individual schedules (this in fact makes $t_s = 0$ for all schedule blocks that this applies to). This would automatically conform with everyone's ideal schedule, thus making all of them be together and generally in their "ideal" day in the theme park (thus also allowing for e as a whole to be maximized). For instance, the late lunch and dinner are all synchronized and all common activities, as well as riding rides in the morning, therefore those are automatically part of the schedule. Since these are curious teenagers who want to try out as much as possible, the shows are also all synchronized in the schedule.

From there, we would take schedules that no one disagrees on, meaning that even if it is not on the exact time slot, all of the friend group really agrees on doing a certain activity, then it would be added to a certain time slot. In other words, if all of the friend group wants to do a certain activity, then we would place it on the most-agreed time slot (the block of time in which most of the friends agree that the activity should be placed). For instance, the games are scattered throughout though after considering the surprisingly many intersections in the schedule, we have been able to fill the games in the respective spots which were also the spots of the other people.

Then finally make some compromises by either getting the majority of the friend group to go to a certain activity, or split up the friend group as they are willing to do so for at most four hours. We would also finetune heavily based on what we think looks "better" as a schedule for the theme park. This is what would be discussed more of in the next section, 5.3: Final Schedule. This is more of a trial-and-error and check-and-balance approach that is brute force, however still keeping in mind only certain choices as to optimize the brute force that we are going to perform.

5.3 Final Schedule

For the final schedule, we decided to add a dinner time after further contemplation of the group's needs. We placed all of the rides at the start of the day (before 12:00pm) because that is when the waiting times are the shortest and the rides don't interfere with Trapezoid Show viewing time. The morning is also when we can assume the group members are the most alert and energetic, giving them energy to mass ride all the coasters (and others). We took intersecting parts from each of the individual schedules and settled on the final schedules.

We decided to split the schedule into two groups: Ishmael and Freya, and Karine and Ming, due to Ishmael and Ming's opposite interests. Ming gets motion sick, so Ming only goes on basic and medium thrill rides, while Ishmael is eager to try all the rides, even the ones with extreme thrill levels. Freya also wants to go on all the rides, but especially the water rides, which can fall under the category of extreme thrill rides.

Lunch is at 2:00pm because during that time, the group will not have to wait at all to eat lunch, decreasing t_{w6} (increasing e_K). Dinner is at 7:00pm, leaving time for the Firework show at the end of the day.

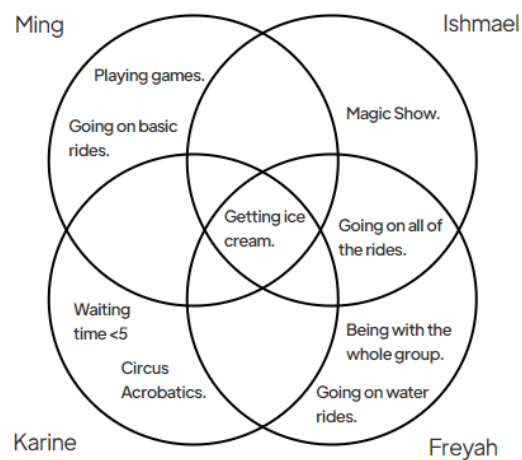
To take the "intersections" of the individual schedules, we created a chart that shows what each of the group members enjoy. We then used a Venn diagram to better represent the data of each friends' ideal schedule/day at the park, and used that to identify what to include in the final schedule.

5.3.1 List of factors for enjoyment in the final schedule

Factor	Increase	Notes
Going on all the rides	e_I, e_F	e_I increases significantly e_F increases quite a bit
Playing games/winning prizes	e_M	e_M increases significantly
Getting ice cream	e_I, e_M, e_K, e_F	They generally like iced cream
Being with the whole group	e_F	e_F increases significantly
Circus Acrobatics Show	e_K	
Magic Show	e_I	

Minimal Waiting Time (tolerance for "no waiting time" is less than 5 mins)	e_K	
Going on basic rides	e_M	
Going on water rides	e_F	e_F increases significantly

The Venn diagram is shown below.



5.3.2 Combined schedule of Ishmael and Freya

Timeslot	Activity
9:00am - 11:02am (Freya & Ishmael)	All the rides (122 minutes) In this order: Decagon Coaster, Heptagon Coaster, Octagon Flume, Rhombus Ride, Starburst Carousel, Concave Train (these 6 first because they already have waiting times at 10:00am) = 32.5 minutes + $t_g(6) = 74.5$ Dodecagon Rapids, Hexa-Swings, Pentagon Bumpers, Square Scramble (these 4 because they don't have waiting times at 10:00am) = 19.5 minutes + $t_g(4) = 47.5$ $74.5 + 47.5 = 122$ minutes

	The walking time can be their breaks in between rides.
11:02am - 11:30am	Convene with Karine and Ming and break time $t_g + 20$ minutes break ≈ 28 minutes
11:30am - 1:00pm	Games $t_g + 75$ minutes playing + $t_g \approx 89$ minutes ≈ 90 minutes
1:00pm - 2:00pm	Miscellaneous Restroom (WC) ≈ 1 hour
2:00pm - 3:00pm	Late lunch @ Triangle Restaurant $t_g + 50$ minutes eating ≈ 57 minutes ≈ 60 minutes
3:00pm - 4:00pm	Ice cream and restroom (WC) $t_g + 30$ minutes eating + $t_g + 10$ minutes washing up and WC ≈ 54 minutes ≈ 60 minutes
4:00pm - 5:00pm	Circus Acrobatics
5:00pm - 6:00pm	Souvenir Gift Shop (t_g minutes walk, 40 minutes shopping, t_g minutes walk to the Circus Acrobatics show)
6:00pm - 7:00pm	Magic Show
7:00pm - 8:00pm	Dinner @ Triangle Restaurant
8:00pm - 9:00pm	Fireworks

5.3.3 Schedule of Ming and Karine During the Split

Timeslot	Activity
9:00am - 9:32am	Basic rides (Starburst Carousel and Concave Train) $t_g + 3 + t_g + 15 = 32$ mins
9:40am - 10:50am	Games
11:00-11:30	Convene with Ishmael and Freya and break time

6 | Analysis

6.1 Strengths

- We kept in mind each of the group member's interests and preferences when creating the schedules with minimal compromises, based on their description.
- The schedule makes use of every available moment guaranteeing that the group will be able to take full advantage of the activities, whilst placing breaks in between to still make it a lax vacation.
- We added breaks for the friends to go to the restroom (WC) and get snacks. As mentioned before, these friends are people, not robots, and they will need time to rest.
- The schedule is clear and concise and you can see that everyone has equal enjoyment with the schedule that we made (refer to 5.3.3 List of factors for enjoyment in the final schedule).

6.2 Limitations

- Every assumption is a limitation. For example, the walk times are not accurate because we simply calculated the maximum time it would take, not the actual time for each distance.
- We worked on the assumption that all the data given to us was correct and unchanging. Of course, real-life data is never accurate, because things like wait time and eating time vary from person to person. All of the data given to us are the *predictable* outcomes, but sometimes events are unpredictable.
- We used a brute force trial-and-error method to formulate the schedule, which means there is a high chance that there may be a better schedule.

6.3 Reflection

Our schedule would differ in many ways, mostly due to the difference in interests compared to the friend groups. Otherwise, due to our assumptions and our model being a bit based on how we would organize our schedule, it would also turn out to be quite similar to each other in a general sense. For instance, one of our team members, Member A, personally likes hot dogs. This is a stark contrast to the character of Karine, because as the problem states, she does not eat hot dogs.

Member A likes eating hot dogs. Member A also loves to play games, but otherwise generally likes to go with the flow. This character is quite similar to Ming and Freya, so there would not be too much of a difference with the original strategy and techniques we applied to our solution and model.

Member B likes playing games and is fascinated by magic tricks. Member B wants to spend as much time as he can with Member A. This is also another way as to how exactly our schedules may differ compared to the original friend group as though we have more personal preferences rather than generic descriptions that are not too complicated. Member B also is quite

a person with a strict schedule — even during vacations. This makes it even harder to schedule, though we could always apply similar strategies to our solution and ultimately come up with a solution. The thing with Member B however is that Member B is only more of time-constrained, rather than activity-constrained, and therefore we may allow for the schedule to fit Member B's personal clock but not necessarily many activities that are related to them. This is mostly the "compromising fine-tuning" part of the solution, together with the "consensus" part.

Member C wants to buy a souvenir and go on exciting rides. Member C loves ice cream. Member C wants to ride water rides to be refreshed. Member C loves watching magic tricks as well. This is quite similar to the usual constraints that Freya presented, so therefore its not that far off from the schedules.

Member D dabbles in circus training so he wants to see the Acrobatics Show. Member D likes to eat big lunches and dinners with his friends. Member D's dream is to run a clown show because he likes making people laugh. Member D also gets quite scared in extreme roller coasters. This is also quite similar to Ming's and Karine's likes and dislikes.

Those would be the statements we would base our personal schedule on. It wouldn't be completely different from that of Ming, Ishmael, Freya, and Karine's schedule, but it certainly has some differences and personality clashes. The reason it would be kind of similar to the group's schedule is because there are very few things to actually like or dislike (since the amusement park is quite small, according to the problem statement), so there would be at least one person who likes the same things as Ishmael or the others. An example of this is Member C, her interests are very near the interests of Freya. However, on the same note, there are still quite a selection of different sites in the theme park, as well as many different characteristics a human could have; these circumstances complicate the problem further.

7 | Conclusion

The task that we accomplished was to create a schedule that would allow for the group of friends to have a very relaxing and enjoyable day in the theme park. We decided to break down the problem by creating an optimal schedule for each person based on their description, then from there join all of these schedules to create a very optimized schedule. Our solution is generally effective, as not only does this create an optimal schedule for the friend group, but it is also very adjustable with different time constraints, and even conforms well with different people with different characteristics and preferences. To make this schedule even better, a possible proposal would be to explore more possibilities and remove certain assumptions that are quite needed when planning for a very optimal schedule for a theme park schedule vacation.

Overall, we carried out the problem successfully, efficiently, and we think we made one of, if not the best possible schedule possible for our group of friends. We are sure that they will enjoy their day and more at Polygon Paradise Park.

8 | Letter to Group

Dear Ming, Ishmael, Karine, and Freya

We are happy to inform you that we have completed and refined our schedule for your upcoming trip to Polygon Paradise Park. It is attached to this letter below. In the making of this schedule, we focused on your personal interests and tried to include everything, and we did that successfully. We went in depth in our research of the park, and regular human needs in a 12-hour period. We also put an emphasis on flexibility, because one of the limitations of our schedule is that the data we got from Polygon Paradise Park is not guaranteed.

Here's what our schedule offers:

- Allotted time for all the activities available in Polygon Paradise Park.
- Lunch time and Dinner time.
- Short waiting times and breaks to rest or repeat activities should you want to.
- Allotted time for going to the restroom (WC) and getting snacks.
- Lenient schedule blocks to account for changes in situation or unpredictable outcomes in the schedule such as different waiting times, eating times, and restroom duration.
- Time to go to the Souvenir shop and the Trapezoid show.
- Time to attend events that certain of you guys like certain shows and activities.

With this, we must explain what this means. We suggest you split up into groups to match your interests and be able to focus on activities that you actually want to do. We've decided that we would fit most rides in the morning, specifically the 9:00am - 11:00am timeslots, seeing as these are the times where there are no lines (for maximum enjoyment!). Then we've allotted the timeslots 11:00am - 11:30 for regrouping and allowance just in case the data we've received about the rides are varied since the data is unpredictable.

Next we've allotted the timeslots of 11:30 - 12:30 for exciting games at the arcade to make Ming happy. Now for 12:30 - 1:00, we've decided the group should get ice cream/snacks, to ease the burden of waiting for late lunch. For 1:00 - 2:00, this will be the time for you to unwind, relax, and go to the restroom (or water closet, however you call it).

Finally, LUNCH! This will take up timeslots 2:00 - 3:00, the reason we postponed lunch was to make Karine happy, who doesn't really like long lines, and they ate ice cream/snacks earlier so they shouldn't be too hungry. Then from 3:00 - 4:00, we've decided to go SHOPPING! During this time, you can check out the souvenir shop and buy mementos that will remind you of your wonderful trip, especially useful for Freya!

The final stretch of your trip to Polygon Paradise Park will start off with watching the Circus Acrobatics show, which we're sure Karine will enjoy! Afterwards, you can head to the hotdog stand and snack on some juicy dogs, and while we recognize that Karine does not eat hotdogs, we suggest Karine saves some leftovers from The Triangle Restaurant to eat for a snack. After that, it's time to watch the Magic Show which Ishmael requested to see. Oh, and we're sure by this time you are longing for a nice meal, so head over back to The Triangle Restaurant to eat a big hot dinner from 7:00 - 8:00pm. And finally, end your day with an exciting Firework Show at 8:00pm.

We at MidMCM Travel Agency hope you have a great time at Polygon Paradise Park! It has been a pleasure making this schedule for you and we hope you hire us again for your other scheduling needs.

Sincerely,

Group # 13575 of the MidMCM Travel Agency.

9 | References

- Inspiration on the gist of how to make a good theme park schedule:
<https://www.calendar.com/blog/how-to-manage-your-time-at-a-theme-park/>
<https://www.iwillteachyoutoberich.com/blog/how-to-plan-a-vacation/>
<https://www.nomadicmatt.com/travel-blogs/planning-a-trip/>
- Times for WC:
<https://www.healthpartners.com/blog/healthy-poop-chart/#:~:text=How%20long%20it%20takes%20you,%2C%20you're%20probably%20constipated>
- Walking Speed of Person:
<https://www.healthline.com/health/exercise-fitness/average-walking-speed>
- On People and Bandwagon:
<https://effectiviology.com/bandwagon/>